

Federated Learning Using MIMIC-IV Dataset

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Background, Related Work, and Objective

Background

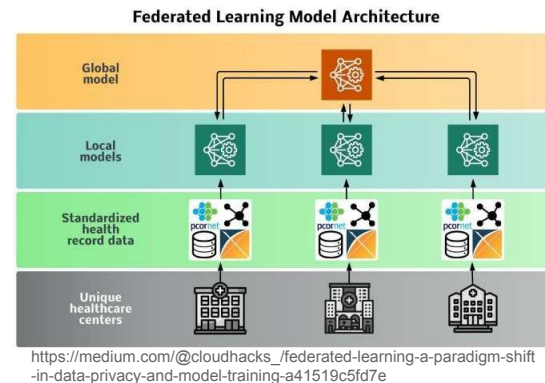
- Patient's medical records are bound by HIPAA and is kept private by the institution
- Legal and security concerns obstructs collaboration between institutions
- Federated learning allows collaboration between institutions without sharing proprietary/confidential information

Related work

- FL Mortality predictor trained on MIMIC-IV dataset achieved accuracy of 88.6%, a recall of 81.7%, and a specificity of 96.5%. [1]
- FL prediction of clinical events from MIMIC-IV dataset with vital signs achieved a classification accuracy of 98.9%. [2]

Objective

- Demonstrate the application of Federated Learning on model training and testing for Length-of-Stay prediction based on the MIMIC-IV dataset.



[1] Vieira, P.; Maia, E.; Praça, I. Privacy-Preserving Mortality Prediction in ICUs Using Federated Learning. In Proceedings of the 18th International Joint Conference on Biomedical Engineering Systems and Technologies (BIOS-TEC 2025) - Volume 2: HEALTHINF, 2025; pp. 87-95. <https://doi.org/10.5220/0013120600003911>

[2] Rakhmiddin, R.; Lee, K. Federated Learning for Clinical Event Classification Using Vital Signs Data. Multimodal Technol. Interact. 2023, 7, 67. <https://doi.org/10.3390/mti7070067>

Data

Dataset

- Mimic-IV v3.1: Large deidentified dataset of patients admitted to the emergency department or an intensive care unit at the Beth Israel Deaconess Medical Center
- Acute Kidney Injury (AKI) Patients Identified by the Following Codes
 - ICD-9: 5845, 5846, 5847, 5848, 5849
 - ICD-10: N17, N170, N171, N172, N178, N179
 - 5849/N179 is the most common (88.4%) AKI code (Acute kidney failure, unspecified)
- Features of Interest
 - Include first_careunit, length of stay, outcome, demographics, lab results, and vital signs
 - With the exception of first_careunit and length of stay, other variables were excluded if missing data exceeds a threshold

Methods

A distributed ML paradigm where model training occurs locally on each client. Raw private patient data never leaves the institution. Only model weights are shared with a central server.

Our FL Pipeline

1. Partition: MIMIC-IV split by ICU unit (MICU, SICU, CCU) each unit acts as an independent “client hospital”.
2. Local Train: Each client trains on its local AKI patient data (demographics, Day-1 labs, vital signs).
3. Aggregate: Central server collects model updates and applies FedAvg weighted average by client dataset size.
4. Broadcast: Updated global model distributed back to clients for the next round.

Why FL for MIMIC-IV? HIPAA restricts cross-institution data sharing. FL enables collaborative model training while keeping patient records local.

Models, Data Splitting, and Evaluation

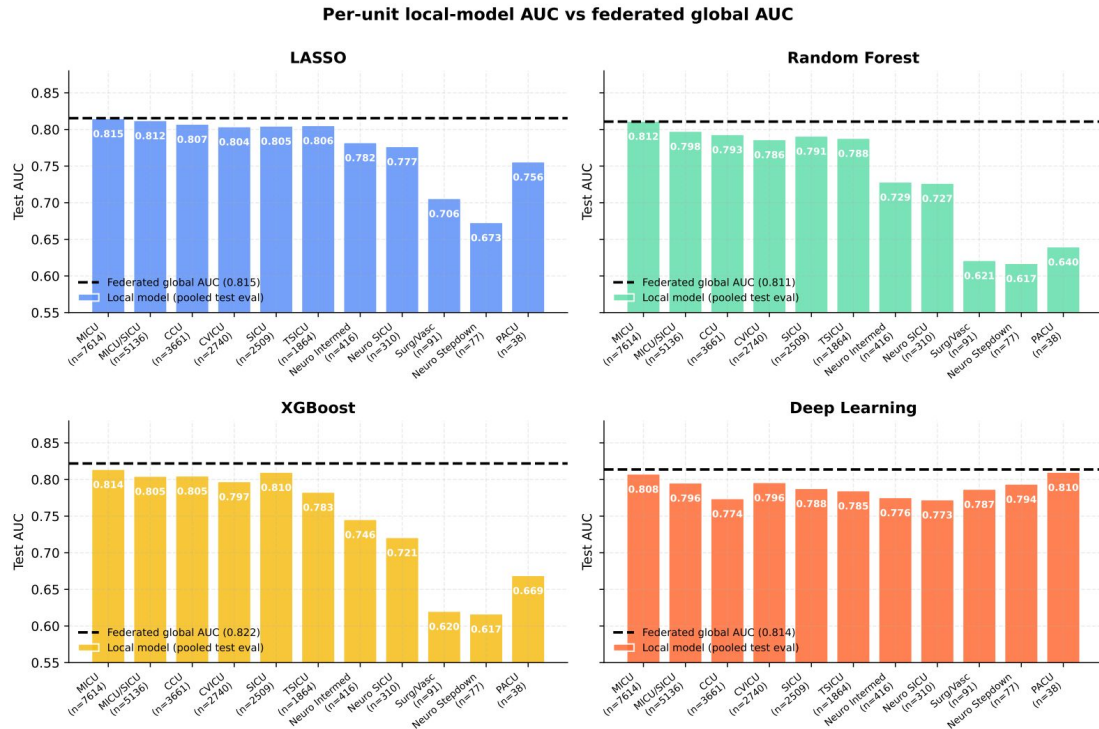
Models: We evaluate four model families of increasing complexity all trained under the same FL framework to assess the performance-complexity tradeoff for ICU Length-of-Stay prediction. These include:

- LASSO Regression (Baseline)
- Federated Random Forest
- XGBoost (Local Ensemble)
- Deep Learning

Data Splitting: The data was split 80/10/10 for training, validation, and evaluation. To ensure that each care unit had enough data to work with after splitting, the data was split for each care unit. Care units with too small or no outcome variation were not used for downstream processing.

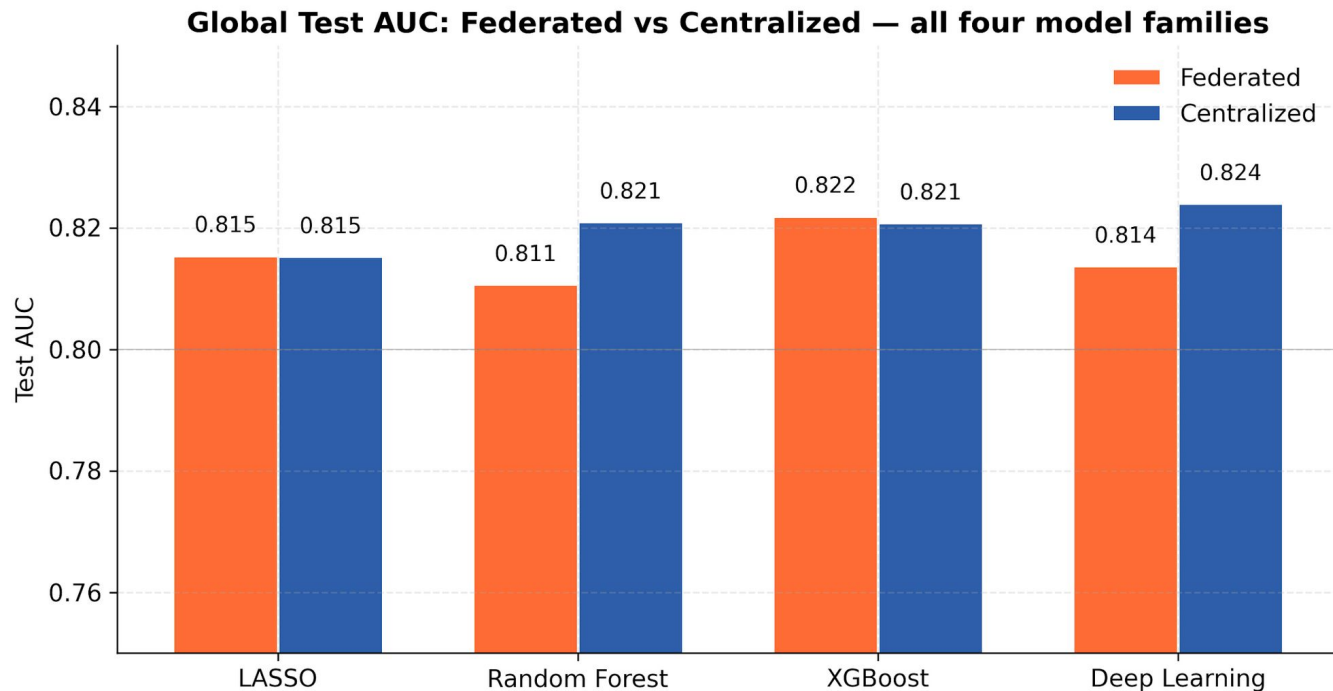
Evaluation: AUC was decided as the key evaluation metric. The model would need to be able to discriminate between short and long length-of-stay, with long length-of-stay being any period of 7 days or longer. Evaluation metrics were collected for all care unit models, federated learning models, and centralized models.

Results: FL outperforms individual care units



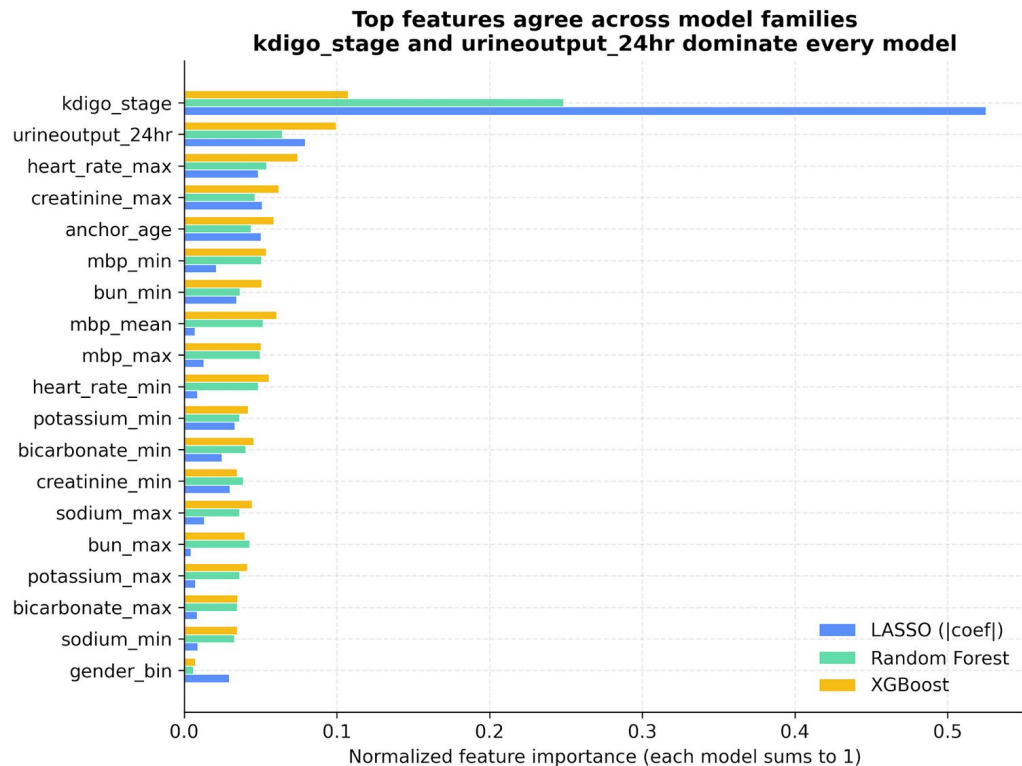
Note: per-unit AUCs evaluate each local model on the pooled test set (all 3,061 patients), not that unit's own test slice. Proper unit-specific evaluation requires re-running with data access.

Results: FL comparable to centralized model



Federated performance is within ~0.01 AUC of centralized for every model family.

Results: Feature importance



Discussion

Findings:

- FL Models outperform most models trained on an individual care unit's data.
- FL Models perform close to centralized models

Limitations:

- FL analysis was done with partitioned data from Beth Israel Deaconess Medical Center and does not represent true cross-institutional heterogeneity
- Variables excluded due to extensive missing data is excluded and may have important information
- Different institutions have unique patient demographics and clinical workflows, a model architecture that excels for one hospital might underperform for another
- Length of Stay (LOS) is not purely a biological outcome; it is heavily influenced by hospital logistics, so it may vary more across actual institutions

Conclusion

- Conclusion
 - FL models are comparable in performance to centralized model, with a maximum AUC decrease of 0.1, allowing for cross-institutional collaboration on ICU Length-of-Stay prediction without transferring protected health information (PHI)
 - Deep learning is the most robust model out of the 4 tested models, with centralized AUC achieving 0.824 and is better generalized
 - Demonstrates that collaboration between institutions is viable without sacrificing much performance
- Possible Improvements
 - Integrate datasets from different sources/institution to mimic the real conditions better
 - Investigating advanced FL aggregation methods (beyond standard FedAvg) to better support poorly performing small-sample units
 - Investigate the susceptibility of shared model gradients to reverse-engineering through Membership Inference Attacks and Data Reconstruction Attacks

Contributions

Vincent: Introduction, data preparation, evaluation, discussion, conclusion

Andrew: Data preparation, methodology, evaluation, discussion

Sina: Methodology, evaluation, results

AI was used to assist in developing and debugging the FL pipeline